

Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages:** Python, SQL, Java(leetcode), C++(leetcode)

• **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Snowflake, PySpark, Workflow Orchestration (Airflow, Step Functions, AWS Glue)

• **Cloud & Tools:** AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD, Docker, GCP, BigQuery, GCS, Spark Streaming

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Built and maintained scalable batch and streaming data pipelines on GCP and AWS, enabling distributed cloud storage and downstream analytics consumption at scale

• Designed and optimized **SQL-based data models and warehouse tables** in PostgreSQL, applying indexing and query tuning techniques to improve performance by 38% for large datasets

• Implemented workflow orchestration using Airflow and AWS Step Functions, adding continuous testing, data quality checks, logging, retries, and failure handling to ensure pipeline reliability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed scalable batch ELT pipelines in Python using GCP BigQuery for high-volume datasets in distributed environments, reducing processing latency by 41%

• Built distributed real-time ingestion pipelines using **Kafka** and **Neo4j**, enabling streaming data processing, transformation, and graph-based analytics at scale

• Designed storage systems and processing frameworks for large-scale distributed data, ensuring consistent availability and supporting evaluation metrics and performance monitoring

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable data processing pipelines using Python, PostgreSQL, and GCP cloud storage layers to support large-scale analytics and downstream data consumption

• Designed fault-tolerant **ETL workflows** for ingesting and transforming high-volume document datasets, ensuring low-latency and reliable data availability

• Developed backend services and used **Git-based CI/CD workflows** to enable reliable deployment and real-time querying across distributed and high-availability environments

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built data pipelines and dashboards to present evaluation metrics and performance KPIs, enabling clear comparisons of improvements and regressions and reducing reporting time by 40%

• Processed large-scale datasets using Python and PySpark for distributed batch processing, implementing transformations, aggregations, and performance monitoring at scale

• Developed **SQL queries** to validate, reconcile, and analyze operational data, improving data quality and speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Built and maintained **SQL-based data pipelines** supporting **Power BI dashboards**, designing data warehouse models, writing complex joins and aggregations, and ensuring delivery of analytics-ready datasets across business functions

• Automated **data transformation workflows** using VBA, implementing validation checks, error handling, and scheduling logic to streamline large-scale data processing and cut reporting time by **42%**

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of **6** engineers to deliver data-driven improvements, analytics insights, and system tracking

• Managed **Agile workflows** using JIRA, tracking delivery, QA issues, and sprint progress to ensure timely and reliable releases

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

• Built an **end-to-end data pipeline** using Python, Pandas, and LightGBM to process large-scale **time-series** transaction data, including ingestion, transformation, feature engineering, and model integration

• Designed and implemented **scalable ETL and feature engineering pipelines** for large datasets, capturing behavioral trends and temporal patterns to improve model performance

• Developed a **data-driven decision framework**, incorporating pipeline outputs and risk thresholds to improve prediction precision by 20% and enable scalable analytics workflows